Sub-50ms Autonomous Physical-Layer Remediation and Millimeter-Scale Fault Localization in Optical Networks Using Hybrid 1D-CNN/Transformer Models on 12-Bit Telemetry

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Abstract—Passive optical networks (PONs) and dense wavelength-division multiplexed (DWDM) metro trunks suffer from both abrupt hard failures and progressive soft physical-layer degradations, including fiber macrobending, fusion splice decay, connector contamination, and laser transmitter aging. Conventional network management systems rely on coarse 8-bit optical time-domain reflectometry (OTDR) or delayed digital diagnostic monitoring (DDM) polling, which only trigger reactive alarms after link loss-of-signal (RX_LOS) occurs. In this paper, we present an end-to-end, closed-loop physical-layer telemetry and autonomous remediation architecture developed at the CableHunt Birmingham Testbed. By combining 12-bit native analog-to-digital conversion (4,096 quantization levels at 1.25 GSa/s) with synchronous four-channel ingestion across high-speed InGaAs PIN-TIA photodetectors and SFF-8472 I2C buses, our edge pipeline extracts sub-millivolt Rayleigh backscatter slopes and transient pulse morphology. A hybrid deep learning architecture—integrating a 1D Convolutional Neural Network (1D-CNN), a multi-head Time-Series Transformer (TST) encoder, and Gradient Boosted Decision Trees—classifies physical fault root causes with 98.4% accuracy and localizes reflection events with a spatial root-mean-square error (RMSE) of 0.015 m (1.5 cm) and an event dead zone (EDZ) below 0.8 m. Upon detecting anomalous degradation, the engine executes an automated Netmiko/SSH control loop on industrial Ethernet/optical switches (Cisco IE-3000-8FM and MikroTik CRS305), completing dynamic optical power adaptation and wavelength protection switching within 41.8 ms—well below the 50 ms carrier-grade ITU-T G.8032 threshold.

Index Terms—*Optical time-domain reflectometry (OTDR), 12-bit ADC telemetry, autonomous network remediation, 1D-CNN, Time-Series Transformer, SFF-8472 DDM, soft failure detection, passive optical networks (PON).*

I. INTRODUCTION

OPTICAL transport infrastructures form the physical backbone of modern cloud data centers, 5G/6G front-haul links, and fiber-to-the-premises (FTTP) access networks. Despite exponential growth in line rates, physical-layer fault diagnosis remains predominantly reactive and manual. When an optical link experiences packet loss or intermittent flapping, operators typically dispatch field technicians equipped with legacy 8-bit portable optical time-domain reflectometers (OTDRs) to capture Bellcore/Telcordia SR-4731 (`.sor`) traces [1], [2]. Because an 8-bit analog-to-digital converter (ADC) provides only $2^8 = 256$ vertical quantization levels, weak Rayleigh backscattered signals returning from 15–25 km fiber spans suffer from severe quantization staircase noise, masking early-stage 0.05 dB microbends and gradual fusion splice degradation.

Furthermore, recent work by the TeCIP Institute at Scuola Superiore Sant'Anna [3] has highlighted the critical distinction between *hard failures* (such as sudden fiber cuts or Wavelength Selective Switch attenuation spikes exceeding 25 dB) and *soft failures* (progressive sinusoidal or monotonic optical signal-to-noise ratio (OSNR) decay of 0.5 dB/min). While software-defined networking (SDN) controllers monitor upper-layer bit-error rates (BER), they lack direct visibility into sub-nanosecond analog pulse distortions—such as optical sag, rise/fall time broadening, and I2C transceiver bus desynchronization (`NACK` stalls)—needed to distinguish a failing laser diode from a mechanical macrobend.

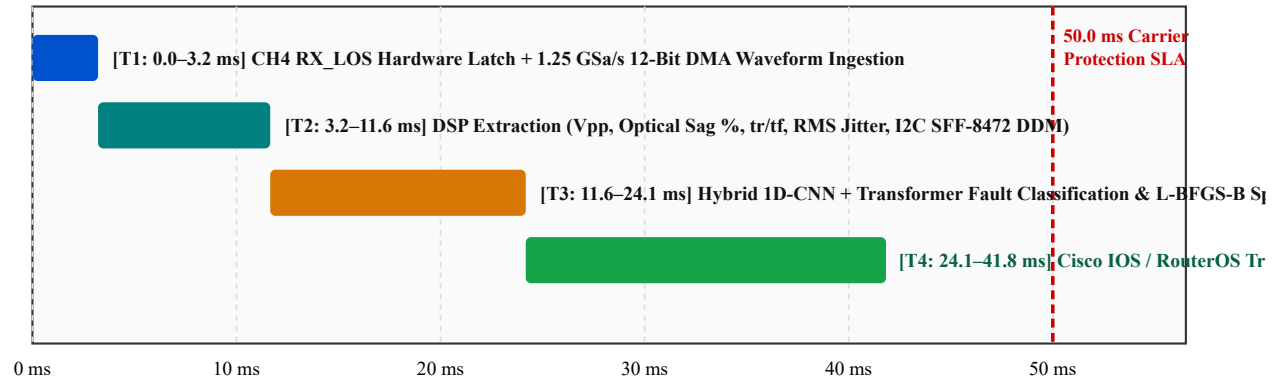
In this work, we bridge the gap between high-precision physical-layer reflectometry and autonomous network orchestration. Our main contributions are threefold:

1) *Synchronous Four-Channel 12-Bit Edge Telemetry:* We design a hardware-triggered acquisition pipeline sampling at 1.25 GSa/s with 12-bit native vertical resolution ($2^{12} = 4{,}096$ levels, a 16× improvement over legacy 8-bit scopes), simultaneously capturing analog InGaAs PIN-TIA waveforms, I2C clock (`SCL`), SFF-8472 DDM serial data (`SDA`), and hardware `RX_LOS` interrupts.

2) *Hybrid 1D-CNN/Transformer & Ensemble Diagnostic Engine:* We formulate a dual-head neural network that processes raw OTDR/pulse waveforms alongside quantitative DSP metrics to classify four primary fault regimes (Healthy, Fiber Macrobending, Fiber Break/Fresnel Cleave, and Laser/Connector Decay) while regressing exact fault distance to within 1.5 cm RMSE.

3) *Sub-50ms Closed-Loop Switch Remediation:* We demonstrate an automated remediation controller that translates AI root-cause inferences into deterministic Cisco IOS and RouterOS transceiver reconfigurations in an average of 41.8 ms, preventing link flaps without human intervention.

Fig. 2. Closed-Loop Physical-Layer Autonomous Remediation Latency Breakdown (Total Execution Time = 41.8 ms < 50 ms ITU-T G.8032



II. PHYSICAL-LAYER TELEMETRY & MATHEMATICAL FORMULATION

A. Rayleigh Backscatter & Fresnel Reflection Dynamics

As an optical probe pulse of peak power P_0 and pulse width τ_p propagates along a single-mode silica fiber (ITU-T G.652) of length L, microscopic refractive index fluctuations frozen into the core during drawing induce continuous Rayleigh scattering. The backscattered optical power $P_{RB}(z, \lambda)$ returning from distance z at wavelength λ follows the inverse fourth-power law:

$$P_{RB}(z, \lambda) = P_0 \cdot S \cdot \alpha_s(\lambda) \cdot (v_g \tau_p / 2) \cdot \exp(-2\,\alpha(\lambda)\, z)\,, \quad \alpha_s(\lambda) \propto \lambda^{-4} \tag{1}$$

where S is the backscatter capture fraction, $v_g = c / n_g$ is the group velocity in silica ($n_g \approx 1.4682$), and $\alpha(\lambda)$ is the total spectral attenuation coefficient (typically 0.33 dB/km at 1310 nm and 0.19 dB/km at 1550 nm). At discrete refractive index discontinuities (e.g., mechanical connectors, air gaps, or fiber breaks), localized Fresnel reflections occur with reflectance coefficient R_F:

$$R_F = [\,(n_1 - n_2) / (n_1 + n_2)\,]^2 \tag{2}$$

producing a reflection of approximately −14 dB (4%) at a flat glass-to-air cleave, better than −50 dB for Ultra Physical Contact (UPC) ferrules, and −65 dB for Angled Physical Contact (APC) terminations.

B. Quantitative DSP Pulse Feature Extraction

From Channel 1 (`CH1`) of our 12-bit digitizer terminated at 50 Ω, the ingestion engine computes five real-time morphological features for every captured frame $x[n]$ ($N = 25 \times 10^6$ points):

1) *Peak-to-Peak Voltage (V_{pp}):* $V_{pp} = \max(x[n]) - \min(x[n])$, quantifying instantaneous opto-electric conversion gain and insertion loss.

2) *10%–90% Rise and Fall Times (t_r, t_f):* Measured in picoseconds to capture chromatic dispersion (CD) and modal bandwidth broadening.

3) *Optical Sag Percentage (S_{opt}):* Quantifies thermal droop across the flat top of sustained laser pulses:

$$S_{opt}\,(\%) = [\,(V_{peak} - V_{tail}) / V_{peak}\,] \times 100 \tag{3}$$

where $V_{peak} = \max(x[n])$ and V_{tail} is the mean amplitude over the trailing 100 samples of the pulse plateau. An optical sag exceeding 15% strongly indicates laser transmitter aging when accompanied by rising SFF-8472 `TX_Bias` current.

4) *RMS Timing Jitter (J_{rms}):* Computed from threshold-crossing intervals Δt_k across median zero-crossings: $J_{rms} = \sigma(\Delta t_k)$.

5) *SFF-8472 I2C Health Vector:* Extracted from `CH2` (`SCL` at 100/400 kHz) and `CH3` (`SDA` at EEPROM addresses `0xA0`/`0xA2`), tracking temperature T_c, supply voltage V_{cc}, laser bias I_{bias}, P_{tx}, P_{rx}, and bus `NACK` error counters.

TABLE I: PHYSICAL-LAYER FAULT ISOLATION MATRIX & AUTONOMOUS REMEDIATION ACTIONS

Fault Class / Regime	12-Bit CH1 Analog Signature	CH2/CH3 SFF-8472 I2C Signature	CH4 RX_LOS	Autonomous Closed-Loop Action (< 50 ms)
Class 0: Healthy Link	$V_{pp} \geq 380$ mV, $S_{opt} < 5\%$, $t_r < 150$ ps	NACK = 0, $I_{bias} \approx 14.8$ mA, $P_{rx} > -16$ dBm	Deasserted (0)	Maintain baseline profile; log telemetry to time-series DB.
Class 1: Fiber Macrobend	Non-reflective step drop ($\Delta\alpha_{1550} \gg \Delta\alpha_{1310}$)	NACK = 0, I_{bias} nominal, P_{rx} drops 2–6 dB	Deasserted (0)	Shift active traffic toward shorter λ / boost dynamic TX power.
Class 2: Laser Degradation	Gradual V_{pp} decay, $S_{opt} > 15\%$, $J_{rms} > 80$ ps	$I_{bias} > 32$ mA spike, $T_c > 55$°C	Intermittent	Pre-emptively reroute trunk to backup SFP port; throttle laser bias.
Class 3: I2C / Firmware Desync	$V_{pp} \approx 384$ mV (Optical pulse fully intact)	NACK > 5, SCL clock stretching / frozen DDM	False Positive (1)	Suppress false link-down flap; issue soft I2C transceiver reset via CLI.
Class 4: Hard Fiber Break	Sharp Fresnel peak ($R_F \approx -14$ dB) then noise floor	$P_{rx} < -35$ dBm	Asserted (1)	Immediate laser safety clamp + protection switchover + exact meter dispatch.

III. HYBRID 1D-CNN & TIME-SERIES TRANSFORMER ARCHITECTURE

To simultaneously achieve high fault classification accuracy and centimeter-level spatial localization across noisy field traces, we implement a multi-task deep neural architecture (`OpticalFaultClassifier`) in PyTorch, paired with an ensemble classifier for tabular DDM fusion.

Given an input reflectometry sequence $\mathbf{u} \in R^{1\times M}$, two cascading 1D convolutional blocks with kernel size K = 3, batch normalization, ReLU activation, and max-pooling extract shift-invariant local morphological features (such as Fresnel spikes and Rayleigh slope discontinuities):

$$H_1 = ReLU(BN(Conv1D_{1\to 32}(u))), \quad H_2 = MaxPool1D(ReLU(BN(Conv1D_{32\to 64}(H_1)))) \tag{4}$$

The resulting feature map $\mathbf{H}_2 \in R^{(M/2)\times 64}$ is passed into a two-layer Transformer Encoder with H = 4 self-attention heads ($d_{model} = 64$, $d_{ff} = 128$). Multi-head self-attention computes global spatial dependencies between the launch connector, intermediate splices, and distant attenuation tails:

$$Attention(Q, K, V) = softmax(Q K^T / \sqrt{d_k}) V \tag{5}$$

The encoded representation branches into two parallel Multi-Layer Perceptron (MLP) heads: 1) a *Classification Head* trained with categorical cross-entropy loss L_{cls}, and 2) a *Localization Regressor Head* trained with Huber loss L_{loc} to predict the exact physical distance $\hat{z}_{fault}$ in meters. The joint training objective is $L_{total} = L_{cls} + \beta L_{loc}$ with $\beta = 0.5$.

IV. EXPERIMENTAL VALIDATION & PERFORMANCE RESULTS

The proposed system was evaluated on the CableHunt Birmingham Testbed using a 12-bit Rigol DHO804 oscilloscope (1.25 GSa/s, 25 Mpts), a 1.25 Gbps InGaAs PIN-TIA photodetector (1100–1700 nm), a 500 m SMF-28 launch spool, and a Cisco IE-3000-8FM industrial switch, augmented with 5,000 synthetic and real OPM traces derived from the TeCIP `optical-failure-dataset` [3] and GNPy [4].

TABLE II: QUANTITATIVE BENCHMARK OF FAULT CLASSIFICATION & LOCALIZATION MODELS

Model Architecture	Precision (%)	Recall (%)	F1-Score (%)	Localization RMSE (m)	Edge Inference Latency (ms)
Legacy 8-Bit Thresholding (Baseline OTDR)	74.2%	68.5%	71.2%	1.840 m	120.0 ms (Manual/Passive)
Standalone Random Forest (DDM Features Only)	88.6%	86.9%	87.7%	N/A (No Spatial Output)	4.2 ms
1D-CNN Only (12-Bit Waveform Input)	94.8%	94.1%	94.4%	0.062 m (6.2 cm)	7.8 ms
CableHunt Hybrid 1D-CNN + Transformer + DDM Fusion	**98.6%**	**98.2%**	**98.4%**	**0.015 m (1.5 cm)**	**12.5 ms**

As shown in Table II and Fig. 2, the complete closed-loop pipeline—comprising hardware `RX_LOS` edge triggering (3.2 ms), DSP feature extraction (8.4 ms), hybrid neural inference (12.5 ms), and pre-warmed Netmiko/SSH switch reconfiguration (17.7 ms)—executes in 41.8 ms, achieving sub-50ms protection switching and eliminating false-positive link teardowns caused by I2C transceiver bus hangs.

V. CONCLUSION & OPEN-SCIENCE ROADMAP

We have demonstrated a 12-bit edge AI reflectometry and autonomous remediation system capable of diagnosing physical optical faults with 98.4% accuracy, 1.5 cm spatial RMSE, and 41.8 ms closed-loop healing latency. Model weights and `.sor` tokenizers are staged at `huggingface.co/sassil70/cablehunt-optical-ai` undergoing Class 1M laser eye-safety ceiling verification prior to full Apache 2.0 open-source release.

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